

Comparative anatomy of wild birds and Fish

Respiratory system of fish

By

Dr/ Basma Gomaa

Assistant professor of Anatomy and Embryology



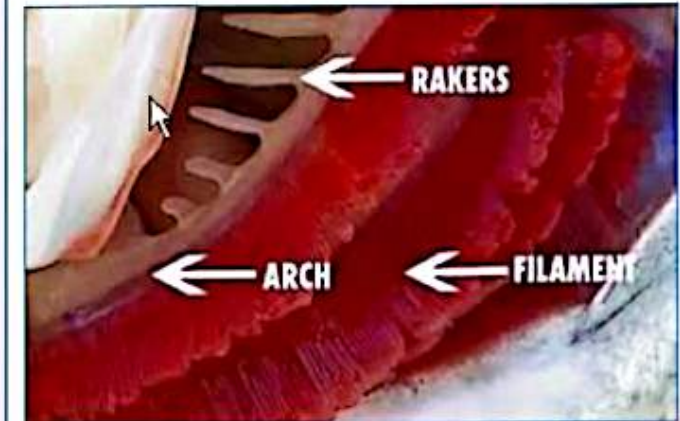
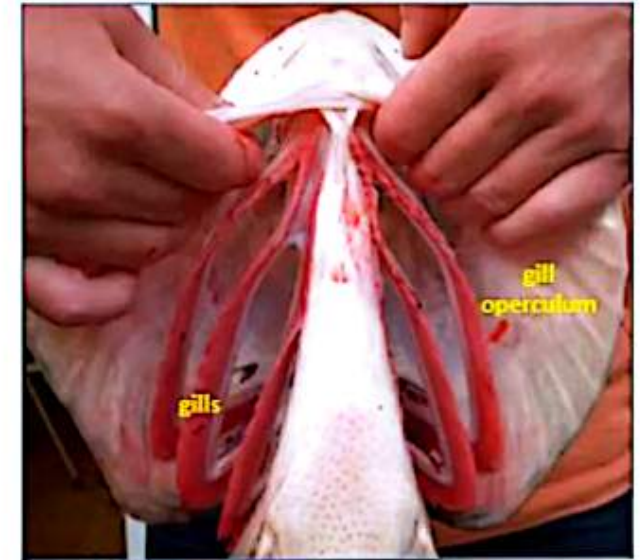
- All aquatic animals without lung used dissolved oxygen in water for respiratory purposes.
- Fish respiration occurs by gills (except the lung fish)
- The gills extract oxygen from water and release CO_2 & other nitrogenous waste products (ammonia)

Function of the gills

- ✓ **Respiratory function:** extract dissolved oxygen from water and expel carbon dioxide
- ✓ **Osmoregulation function:** contain special *chloride cells* that actively excrete or absorb salts (Na^+ , Cl^-) to maintain the body's osmotic balance
- ✓ **Excretion function:** mainly nitrogenous waste products such as ammonia

Gills (Branchia)

- The healthy gills are bright red & donot stick together
- Most teleosts have four pairs of gills covered by the gill operculum.
- The gills locates between buccal cavity & opercular cavity.
- The gills composed of:
 - 1) Gill arch
 - 2) Gill rakers
 - 3) Gill filaments (lamellae)
 - 4) Gill musculature



Gill arches (gill bars)

- It is bony or cartilaginous curved plates on both sides of a fish's pharynx that support the gill filaments and rakers
- **Its cranial border** is concave attaches to gill rakers
- **Its caudal border** is convex attaches to gill filaments and gill rays (**support gill filaments in cartilaginous fish**)
- **Teleosts gill arch formed of 5 bones:**
 1. basibranchial
 2. Hypobranchial
 3. Ceratobranchial
 4. Epibranchial
 5. pharyngeobranchial

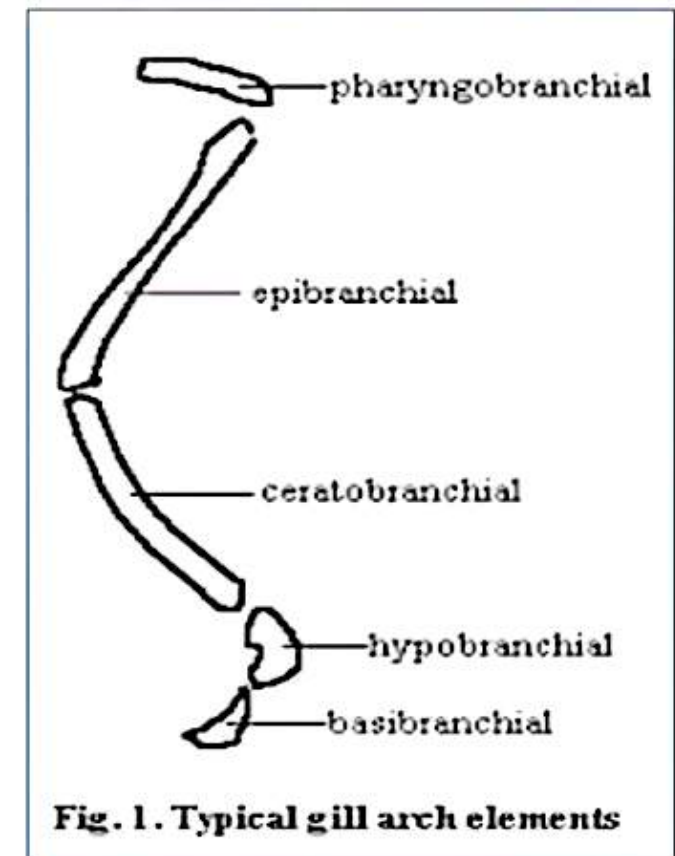
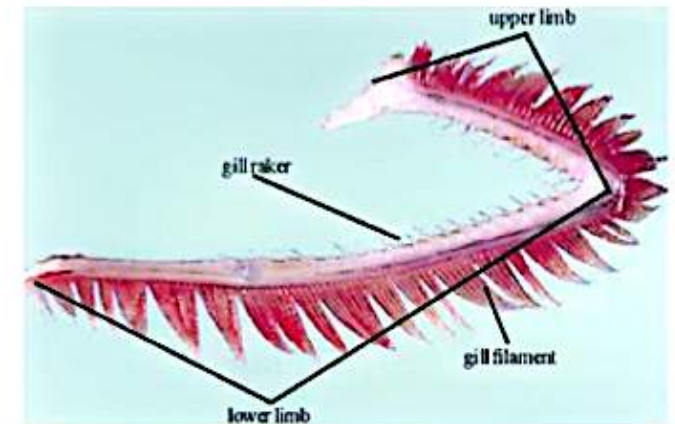
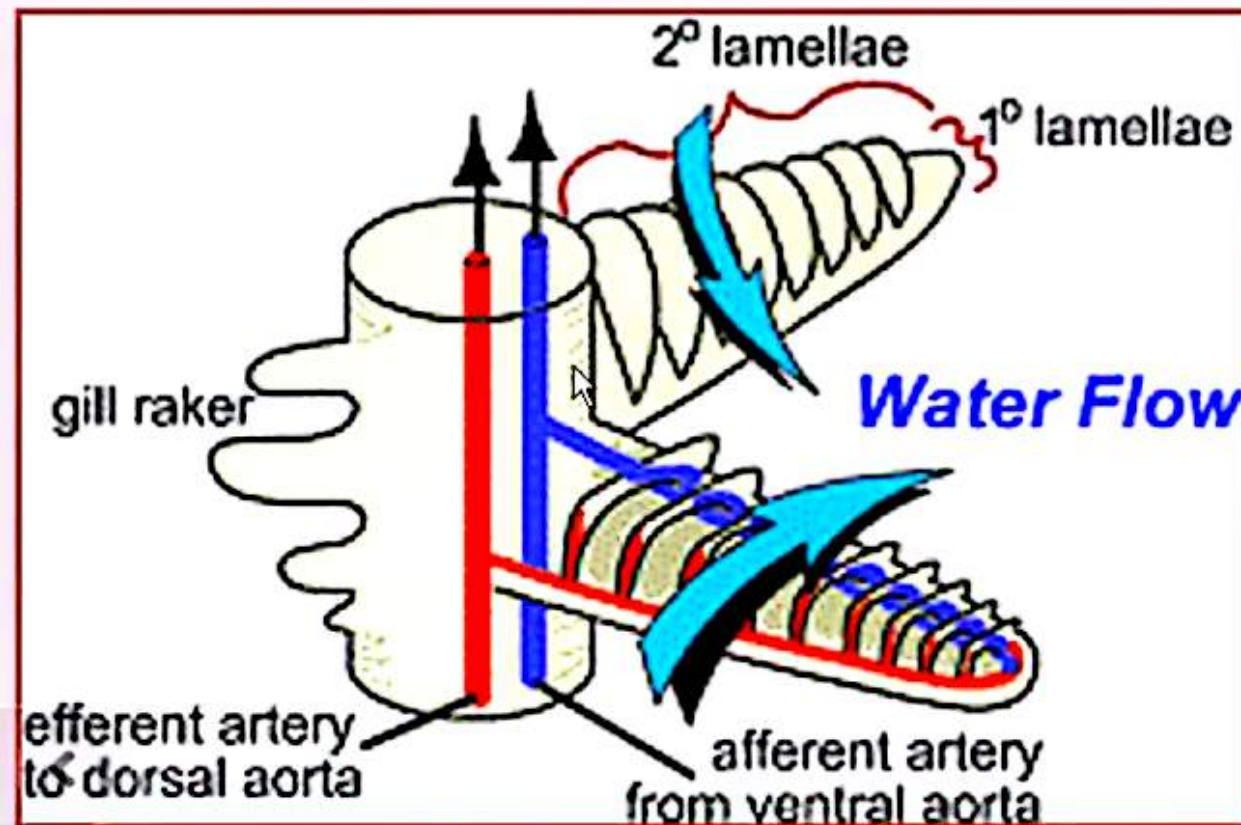


Fig. 1. Typical gill arch elements

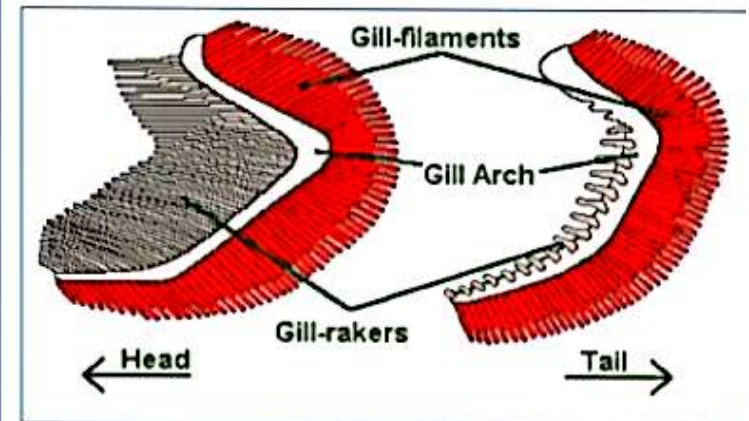
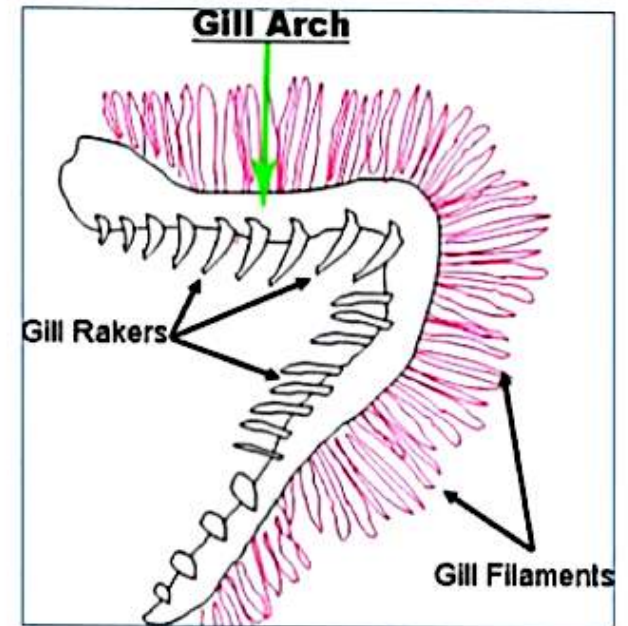
- Within each gill arch, there are two arteries:

1. **Afferent branchial A.:** come from ventral aorta (carries non-oxygenated blood)
2. **Efferent branchial A.:** connect to dorsal aorta (carries oxygenated blood).



Gill rakers

- It is **cartilaginous** or **bony tubercles** directed cranially toward pharynx.
- It **prevent escaping of food particles** from pharynx, so it doesnot pass over gill lamellae.
- Absence of gill rakers or **precipitation of food particles on gill** leading to their **suffocation** of the fish.
- The number & arrangement of the gill rakers vary according to size of food particles:
 - **Plankton Feeders**: Possess long, thin, densely spaced rakers **(like the comb)** to trap small zooplankton floating in the water
 - **Carnivores/Omnivores**: Typically have shorter, sturdier, and more widely spaced rakers



Gill filaments

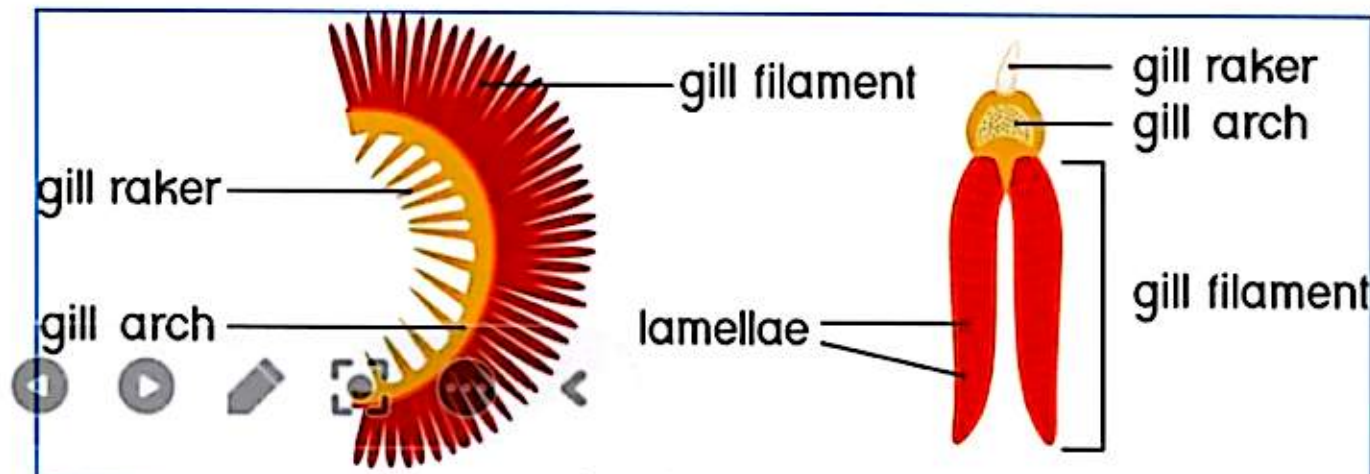
✓ The gill filaments are vascularized structures projected from the caudal (convex) border of gill arch

✓ One holobranch attaches to each gill arch

✓ Each holobranch consists of two hemibranchs (**two rows of gill filaments**)

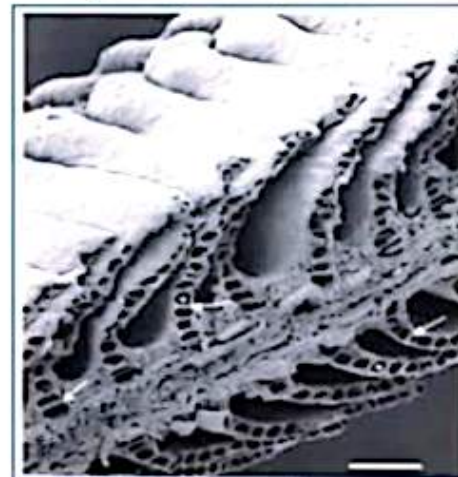
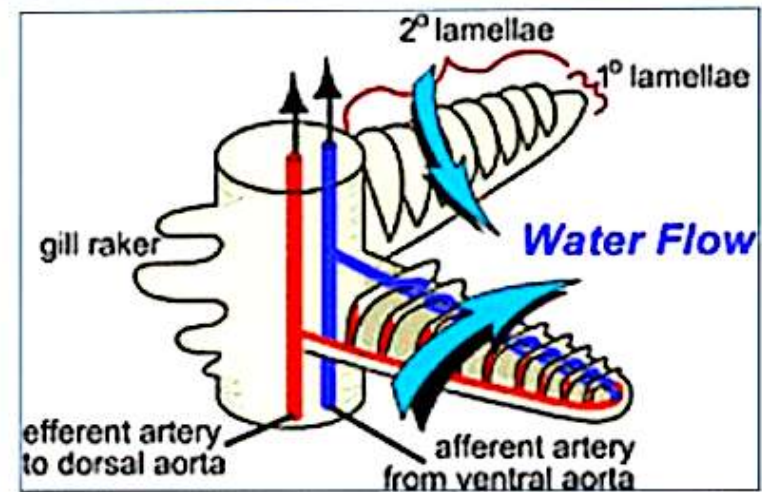
✓ On each side of pharynx, there are:

- 4 gill arches
- 4 holobranches
- 8 hemibranches



- Each **hemibranch** consists of a row of **long thin filaments** called **primary lamellae** which project as teeth of comb.
- The **primary lamellae** carry **secondary lamellae** which project from dorsal & ventral parts of primary lamellae as regular semilunar fold.
- The **secondary gill lamellae** contain within its core numerous wide thin walled capillaries which designated as gill sinusoids (blood spaces) separated from each other by pillar cells which originate from epithelial cells of filament.

- At base of **secondary lamellae** there are special cells called **chloride cells** which are responsible for mineral balance in body.
- secondary filaments are sites of gaseous exchange.



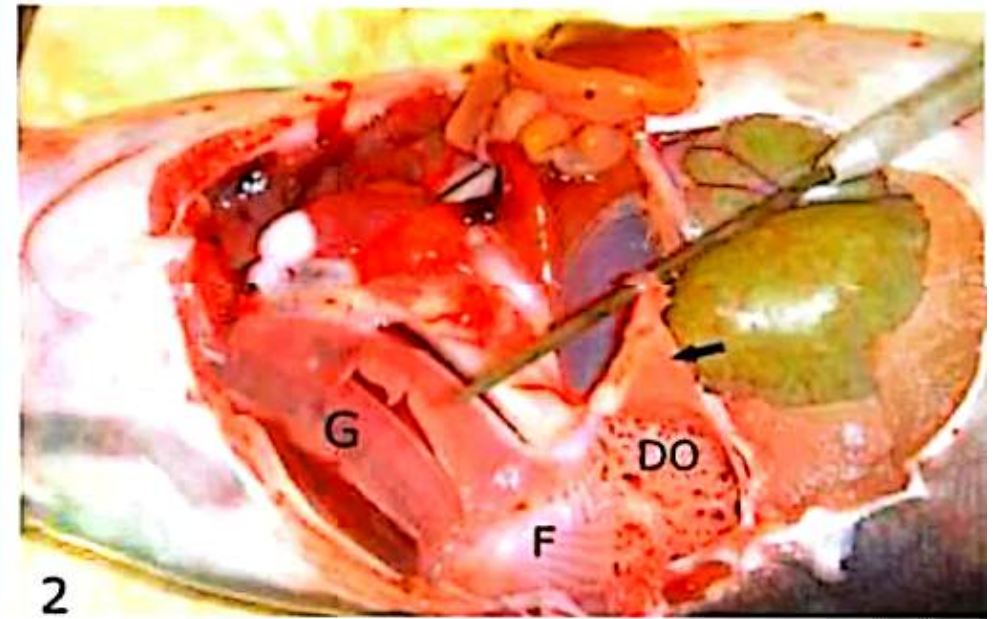
Gills of cartilaginous fish

- Teleosts gills are similar to that of cartilaginous fish, but
 - 1) In most of cartilaginous fish, the number of gills is **5 pairs**
 - 2) in few certain species, the gills are **6 -7 pairs**
 - 3) Each gill is placed in **special cleft** and covered with its **special covering**.



Accessory arborescent (labyrinth) organ
(air-breathing dendretic organ)

- ✓ It Located in suprabranchial chamber caudodorsal to gills
- ✓ It is covered wholly by membrane lining suprabranchial chamber.
- It acts as a **temporary lung** which helps catfish to live for comparatively long time outside water.
- It is consisted of highly vascularized, tree-like (dendritic) extensions of the gill arches

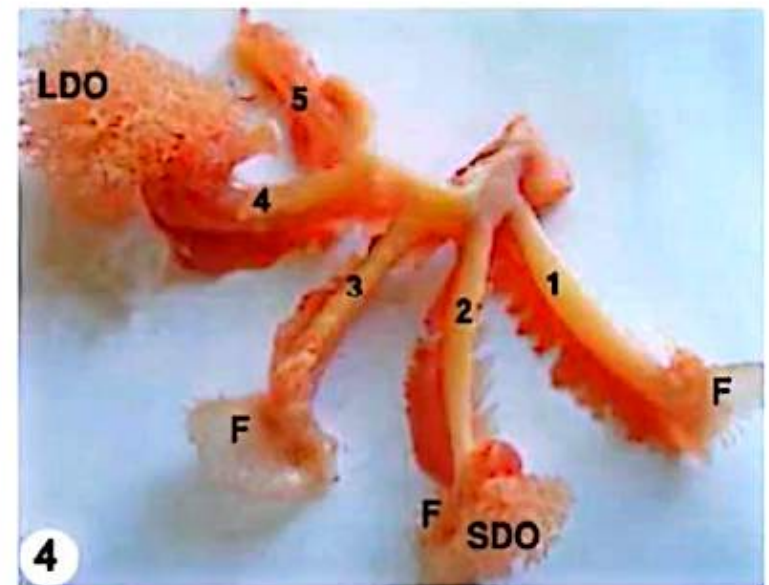
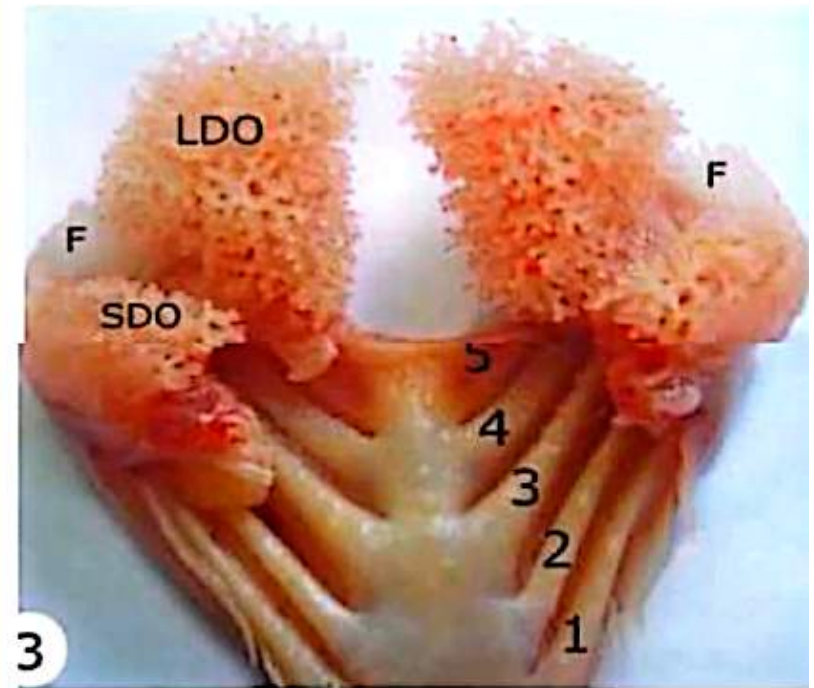


✓ In catfish, the dendretic organ was divided into two parts that originating as out pocketing from proximal end of 2nd & 4th branchial gill arches:

1. small anterior part occupy anterior part of suprabranchial chamber.

2. large posterior part occupy the middle & posterior parts of suprabranchial chamber

✓ Both of the anterior & posterior parts were connected to its corresponding gill arch by **cartilaginous joint**

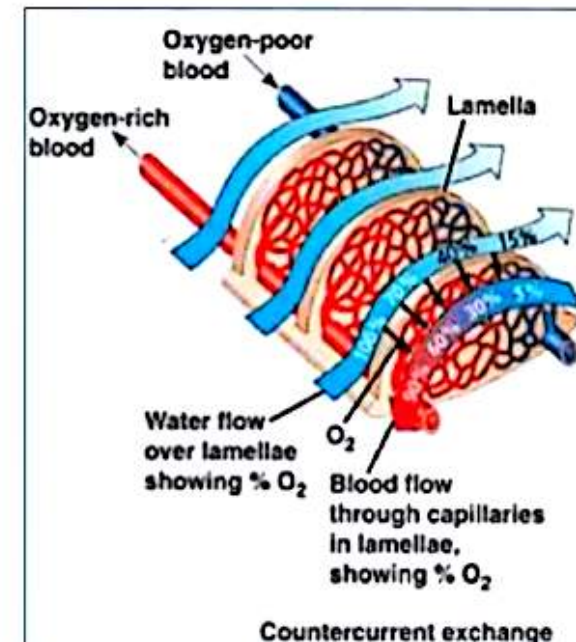


Mechanism of respiration in fish

Counter current flow mechanism

- **The counter-current mechanism in fish** is a highly efficient respiratory process where blood in the gill capillaries flows in the opposite direction to water passing over the gill lamellae
- Blood that is already partially oxygenated encounters water with a higher oxygen concentration, allowing oxygen to continue diffusing into the blood along the entire length of the gill.

- The counter current flow mechanism need a **c o n t i n u o u s** unidirectional flow of water over gill surface.
- This unidirectional flow of water created by **branchial pumping**.

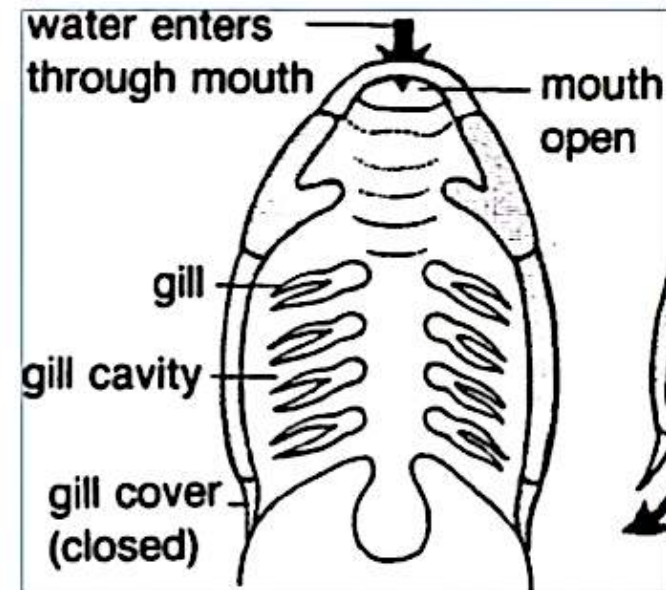


Branchial pumping cycle comprises:

1) Expansion phase: (mouth is opened & operculum valve is closed):

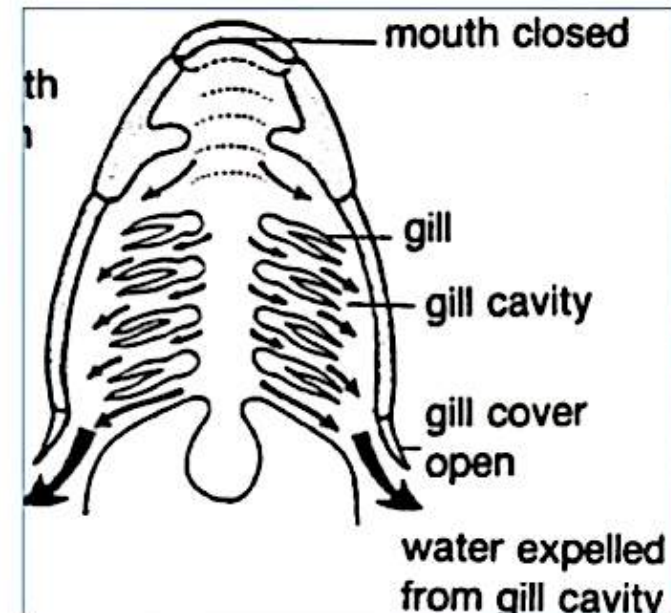
- Water enters mouth and cause expansion of oral cavity.

- Water then passes over gills by a simultaneous contraction of oral cavity & expansion of opercula.



2) Compression phase (mouth is closed & operculum valve is open)

- This make water expelled out of opercular opening.



Swim bladder



- **The swim bladder is** a serous transparent sac contains air.

- It locates at the dorsal part of the body cavity just ventral to vertebral column & kidney.

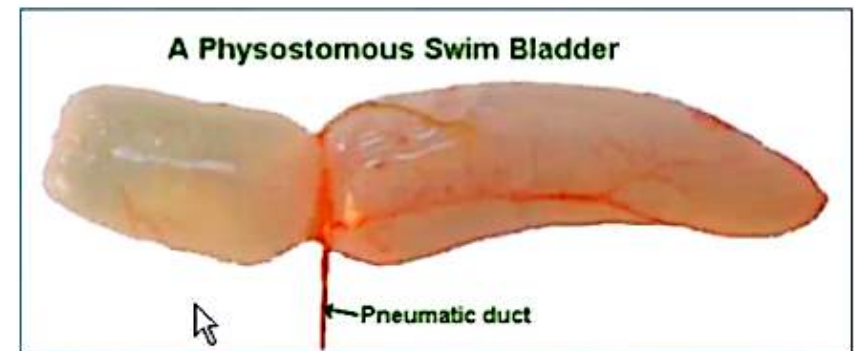
• **Types of swim bladder:**

A) According to connection with esophagus

1. **Physoclistic bladder:** no connection with esophagus as in T.nilotica
2. **Physotomic bladder:** connect with esophagus by pneumatic duct as Carp

B) According to number of chambers

1. **Single bladder:** only one chamber as in Tilapia nilotica
2. **Double bladder:** two chambers separated by a diaphragm and connected by an opening controlled by a sphincter as in Carp.



- ❑ The cranial chamber in Carp & the cranial portion in *Tilapia nilotica* is associated with gas production & consequently has a thicker wall (but smaller size)
- ❑ The posterior chamber/portion is involved in gas reabsorption & has thin wall (but large size)

Function of swim bladder



1- Maintain fish buoyancy (vertical position in water column= suspending= floating).

- It acts just like a balloon- with ability to control amount of gas.

✓ More gas is added to bladder enables fish to move to a higher level in water.

✓ Gas is released from bladder to move to a lower position in water.

- Fish that donot possess a swim bladder sink to bottom if they stop swimming.

2- Sound production: (in some fishes)

* by movement of gases and air through pneumatic duct with help of a powerful extrinsic muscle.

3- Excretory function:

- in herring fish, bladder has a posterior opening to exterior.

